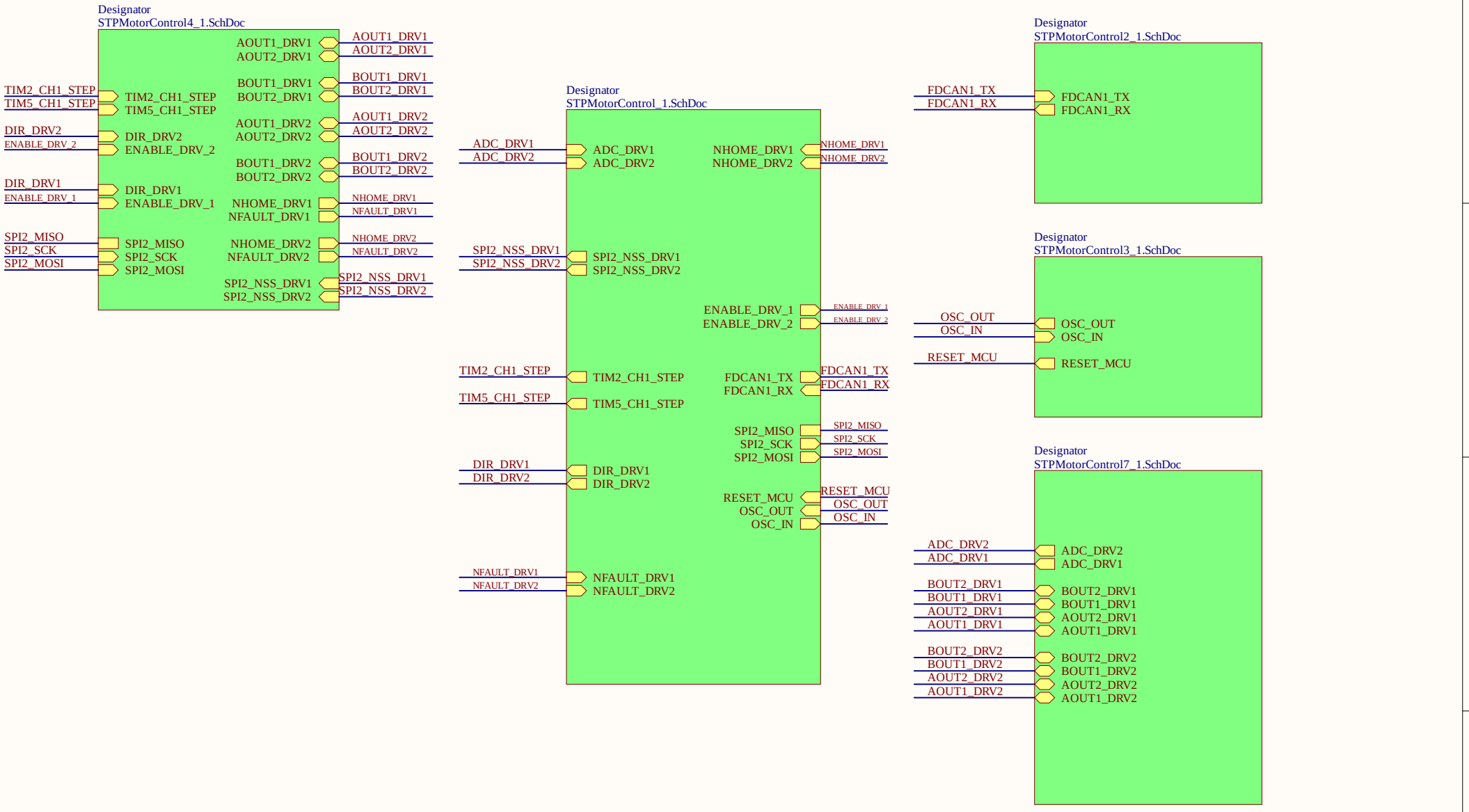
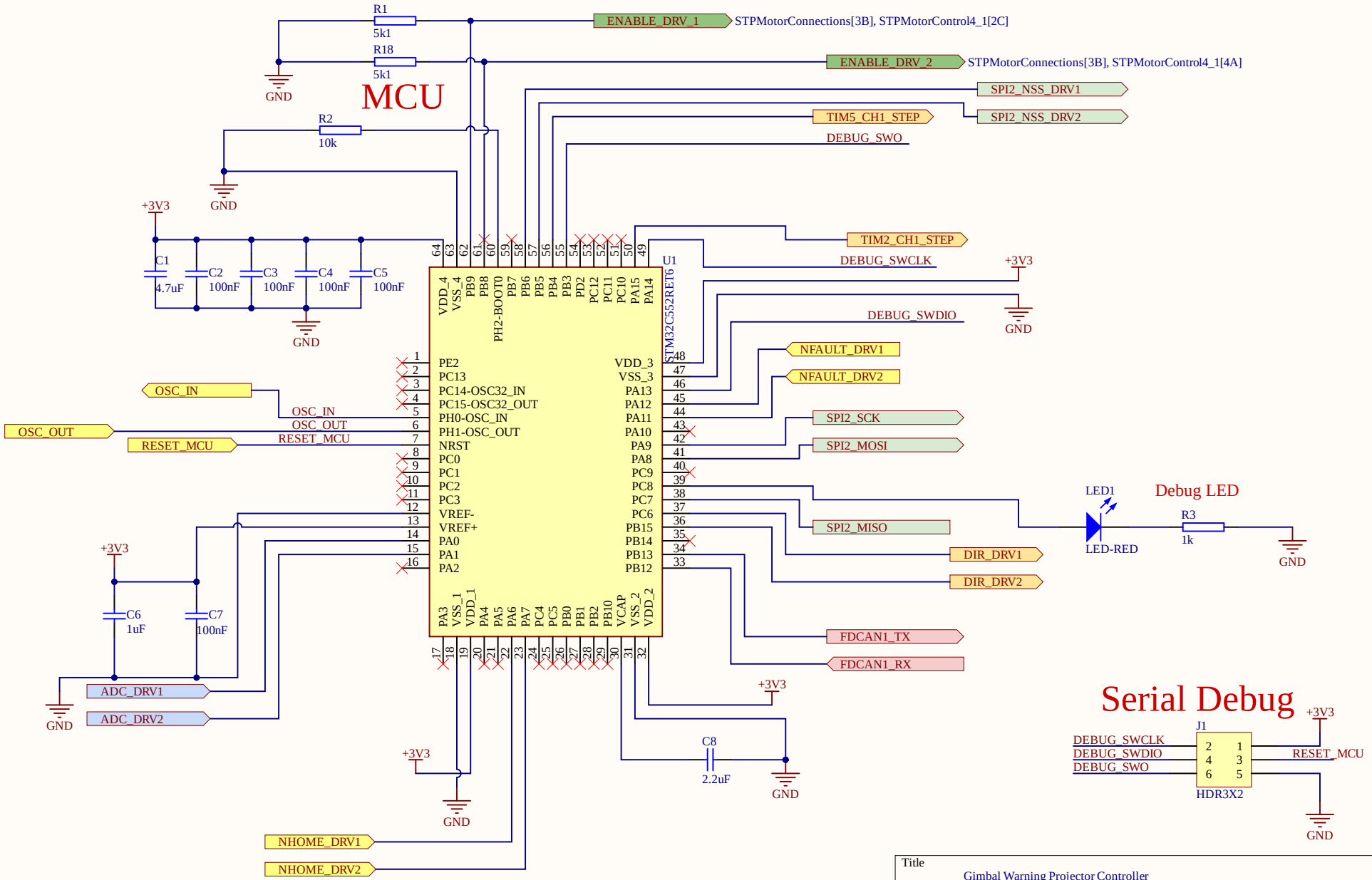
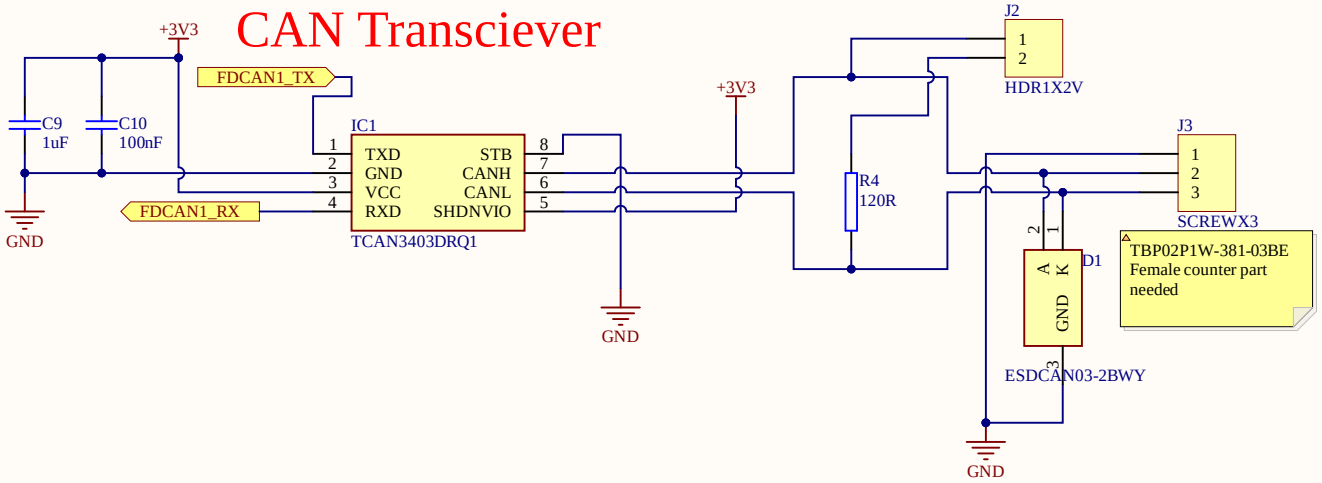


Title	Gimbal Warning Projector Controller	
Size: A4	Number: 1	Revision:
Date: 10/20/2026		Sheet of 1/8
File: C:\Users\..\STPMotorConnections.SchDoc		Drawn By: Mohamed Hanan



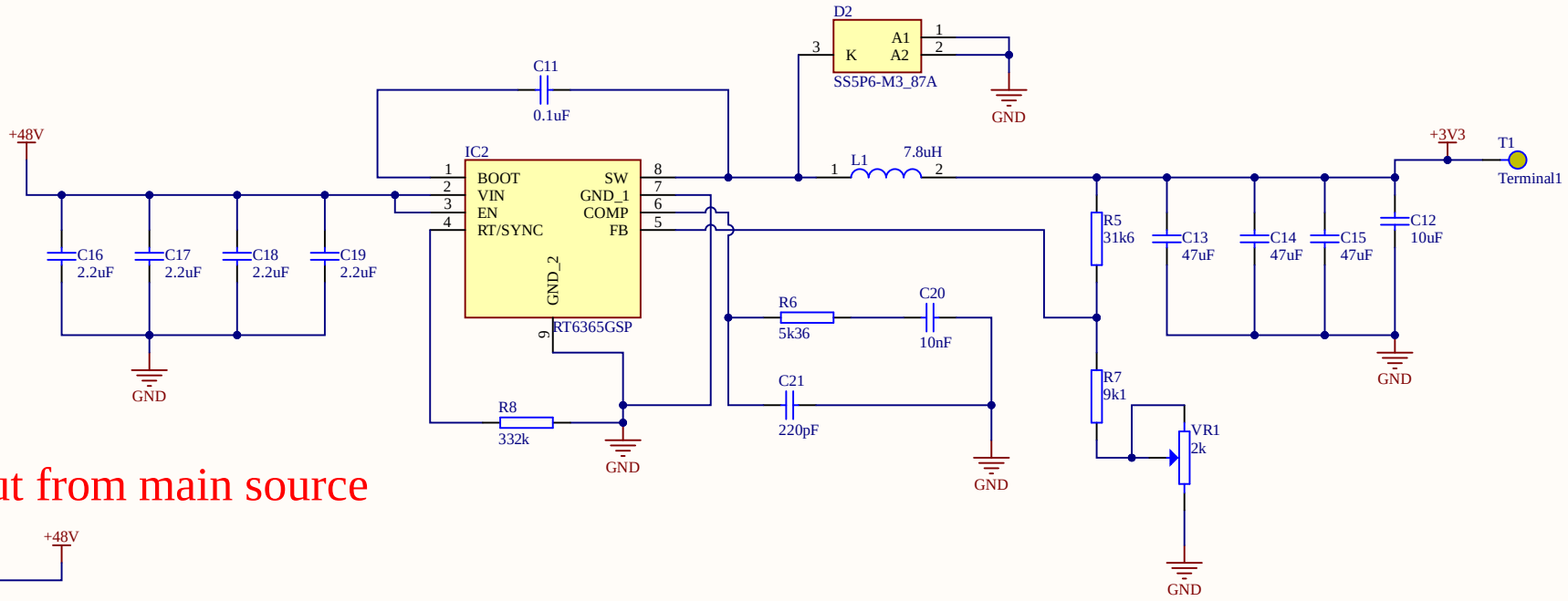
Title	Gimbal Warning Projector Controller	
Size: A4	Number: 2	Revision:
Date: 10/20/2026	Sheet of 2/8	
File: C:\Users\..\STPMotorControl_1.SchDoc	Drawn By: Mohamed Hanan	

CAN Transciever

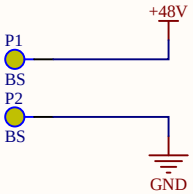


Title	Gimbal Warning Projector Controller		
Size: A4	Number: 3		Revision:
Date: 10/20/2026		Sheet of 3/8	
File: C:\Users\..\STPMotorControl2_1.SchDoc		Drawn By: Mohamed Hanan	

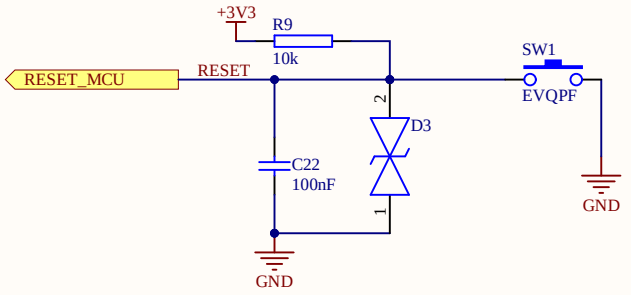
Switch mode converter form 48V to 3.3V



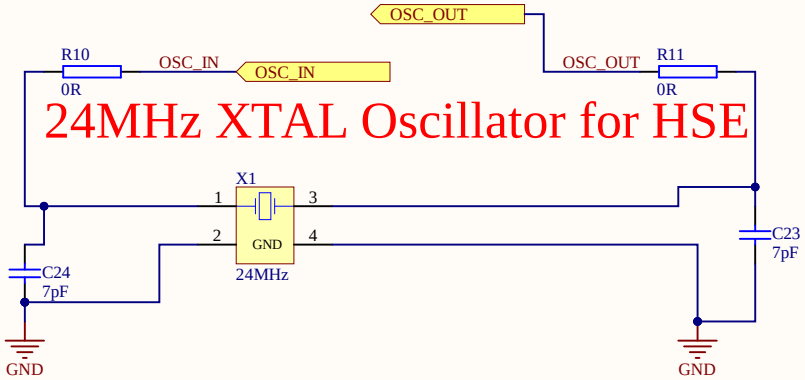
48V input from main source



Switch 4 MCU RESET PIN

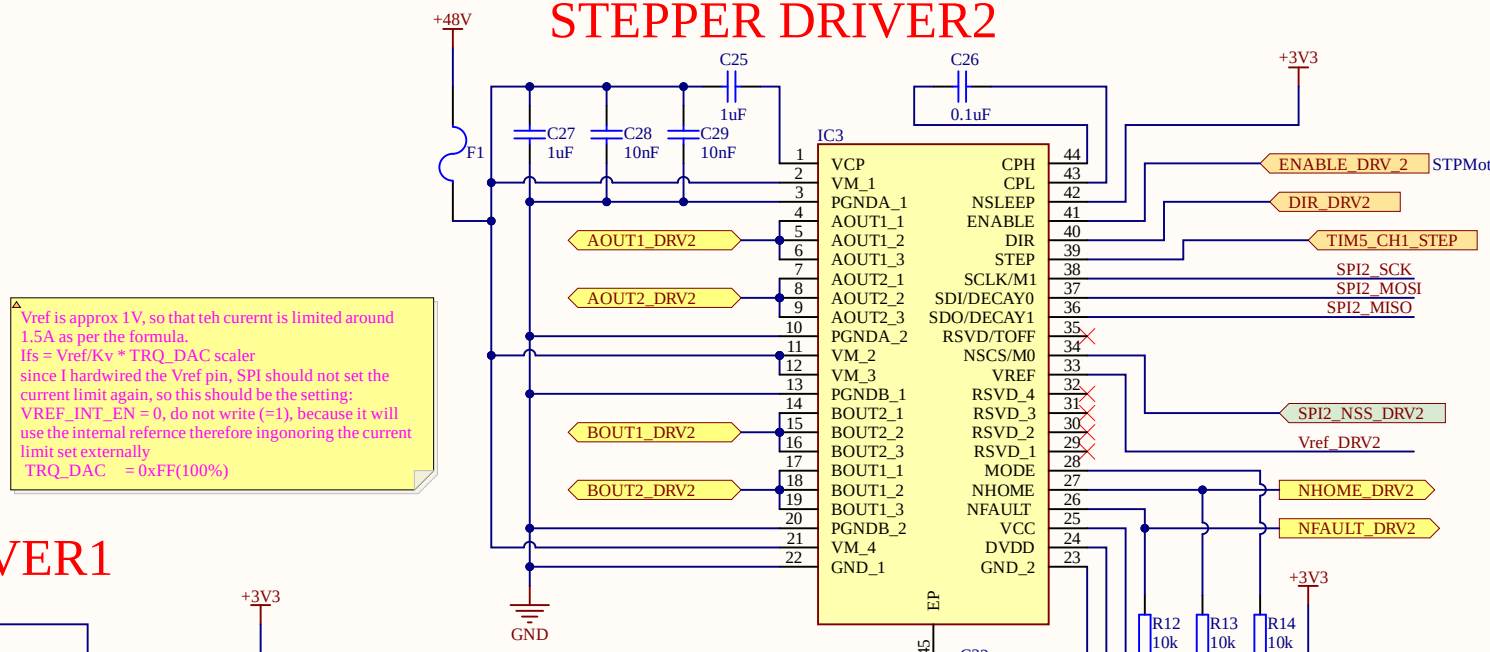


24MHz XTAL Oscillator for HSE



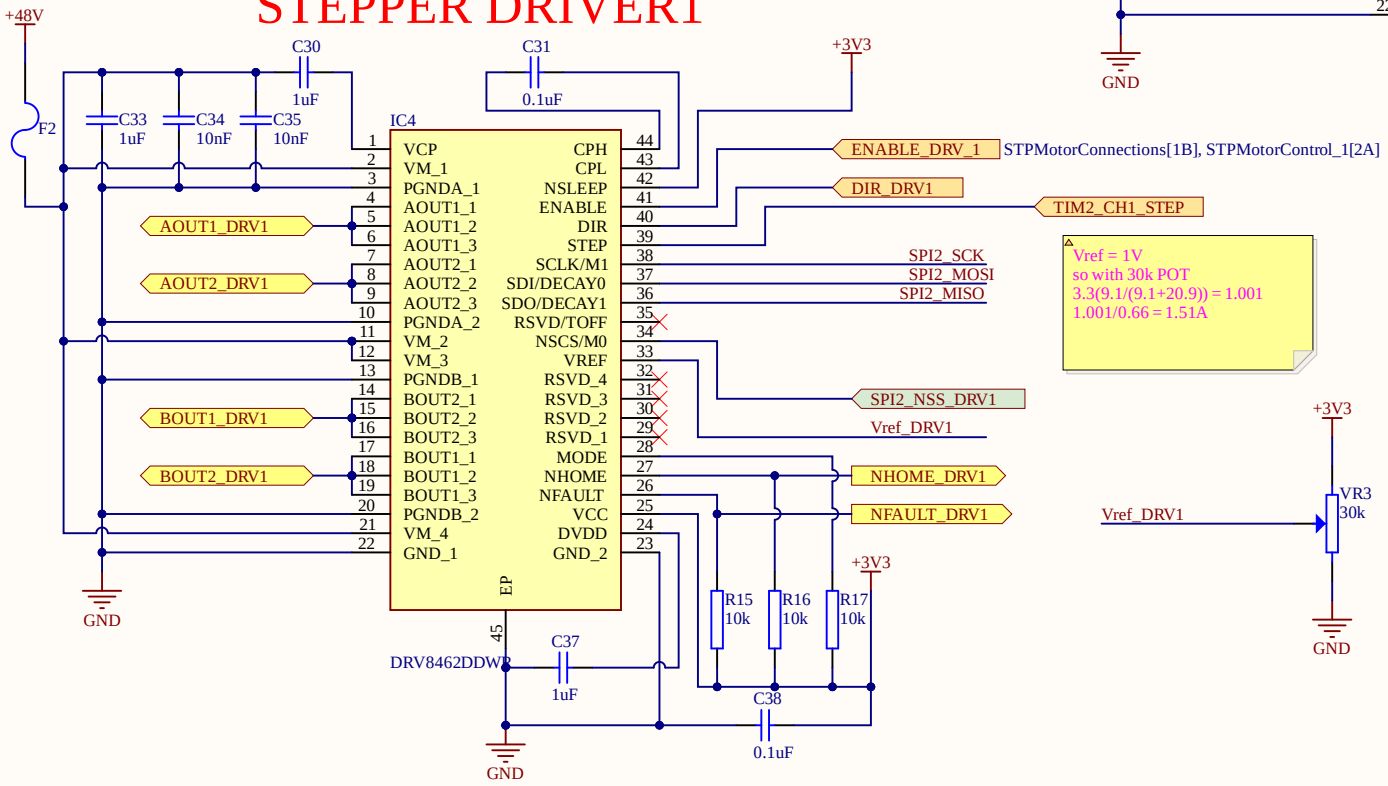
Title: Gimbal Warning Projector Controller		
Size: A4	Number: 4	Revision:
Date: 10/20/2026	Sheet of: 4/8	
File: C:\Users\..\STPMotorControl3_1.SchDoc	Drawn By: Mohamed Hanan	

STEPPER DRIVER2

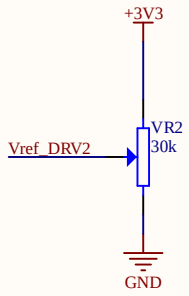


Vref is approx 1V, so that teh curernt is limited around 1.5A as per the formula.
Ifs = Vref/Kv * TRQ_DAC scaler
since I hardwired the Vref pin, SPI should not set the current limit again, so this should be the setting:
VREF_INT_EN = 0, do not write (=1), because it will use the internal refernce therefore ingonoring the current limit set externally
TRQ_DAC = 0xFF(100%)

STEPPER DRIVER1



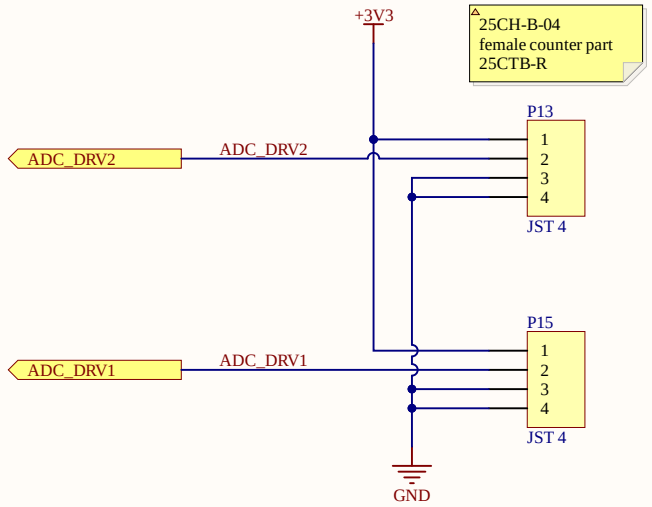
Vref = 1V
so with 30k POT
3.3(9.1/(9.1+20.9)) = 1.001
1.001/0.66 = 1.51A



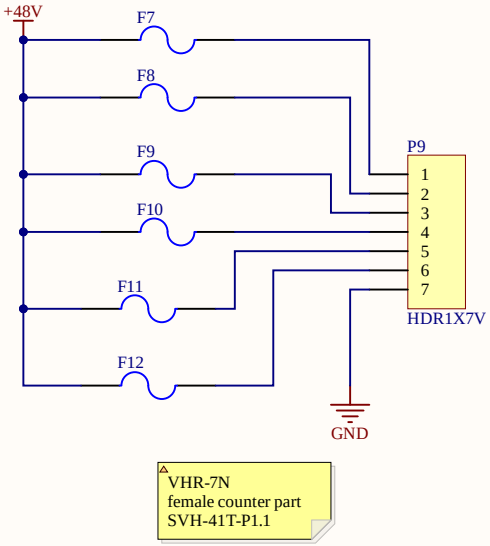
SPI2_SCK
SPI2_MOSI
SPI2_MISO

Title: Gimbal Warning Projector Controller		
Size: A4	Number: 5	Revision:
Date: 10/20/2026	Sheet of 5/8	
File: C:\Users\..\STPMotorControl4_1.SchDoc	Drawn By: Mohamed Hanan	

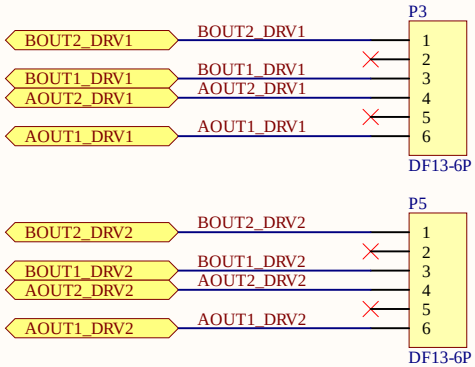
MAGNETIC ENCODER CONNECTOR



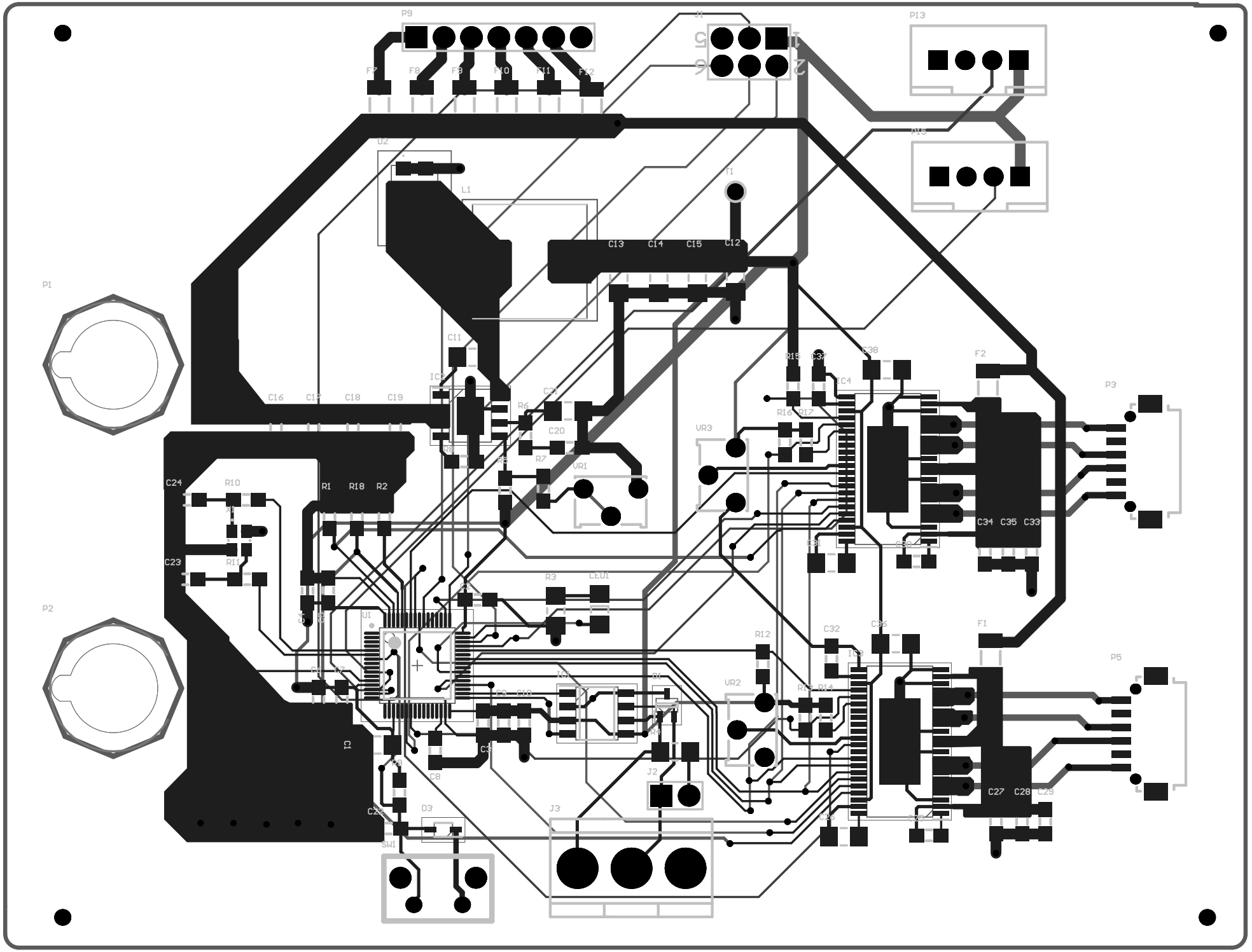
LED modules Connector



STEPPER MOTOR CONNECTOR



Title	Gimbal Warning Projector Controller		
Size: A4	Number: 8		Revision:
Date: 10/20/2026		Sheet of: 8/8	
File: C:\Users\..\STPMotorControl7_1.SchDoc		Drawn By: Mohamed Hanan	


P9
F7
F8
F9
F10
F11
F12
X1
P13
P15
D2
L1
T1
C13
C14
C15
C12
P1
C11
IC2
C16
C17
C18
C19
R6
C21
C20
R7
VR1
R15
C37
C38
F2
IC4
R16
R17
VR3
P3
C24
R10
R1
R18
R2
C34
C35
C33
C23
R11
R3
LED1
P2
U1
R12
C32
C36
F1
IC3
P5
VR2
R13
R14
X2
Q1
C3
C10
C9
C8
J2
C27
C28
C29
C22
D3
J3
SW1

Comment	Description	Designator	Footprint	LibRef	Quantity
4.7uF	Non Polarized Capacitor, SMD, 1206	C1	CAP_1206	CAPN 1206	1
100nF	Non Polarized Capacitor, SMD, 0805	C2, C3, C4, C5, C7, C10, C22	CAP_0805	CAPN 0805	7
1uF	Non Polarized Capacitor, SMD, 0805	C6, C9, C25, C27, C30, C32, C33, C37	CAP_0805	CAPN 0805	8
2.2uF	Non Polarized Capacitor, SMD, 0805	C8, C16, C17, C18, C19	CAP_0805	CAPN 0805	5
0.1uF	Non Polarized Capacitor, SMD, 1206	C11, C26, C31, C36, C38	CAP_1206	CAPN 1206	5
10uF	Non Polarized Capacitor, SMD, 1206	C12	CAP_1206	CAPN 1206	1
47uF	Non Polarized Capacitor, SMD, 1206	C13, C14, C15	CAP_1206	CAPN 1206	3
10nF	Non Polarized Capacitor, SMD, 0805	C20, C28, C29, C34, C35	CAP_0805	CAPN 0805	5
220pF	Non Polarized Capacitor, SMD, 1206	C21	CAP_1206	CAPN 1206	1
7pF	Non Polarized Capacitor, SMD, 0805	C23, C24	CAP_0805	CAPN 0805	2
ESDCAN03-2BWY	TVS Diode (Uni-directional)	D1	SOT65P210X110-3N	ESDCAN03-2BWY	1
SS5P6-M3_87A	Schottky Diode	D2	SS5P6M387A	SS5P6-M3_87A	1
TSD05CDYFR	TVS Diode (Bi-directional)	D3	SOD2613X100N	TSD05CDYFR	1
1206L035/16YR	Fuse, PTC Reset, 350mA Hold, 750mA Tripping, 16v	F1, F2, F7, F8, F9, F10, F11, F12	MC36207	1206L035/16YR	8
TCAN3403DRQ1	Integrated Circuit	IC1	SOIC127P600X175-8N	TCAN3403DRQ1	1
RT6365GSP	Integrated Circuit	IC2	SOIC127P600X175-9N	RT6365GSP	1
DRV8462DDWR	Integrated Circuit	IC3, IC4	SOP64P810X120-45N	DRV8462DDWR	2
HDR3X2	Header, 3-Pin, Dual row, 2.54mm Pitch, Straight	J1	HDR2X3_AUT	HDR3X2	1
HDR1X2V	Header, Plug, 2.54mm, 2 Pins, Straight	J2	HDR1X2_AUT	HDR1X2V	1
SCREWX3	Terminal Block, PCB, 3Way, Pitch 5mm	J3	SCREW3_AUT	SCREWX3	1
7.8uH	Inductor	L1	WE-HCI_103023	744325780	1
LED-RED	RED LED, Case 1206	LED1	LED1206R	SMD 1206 LED-RED	1
BS	Banana Socket, 4mm, Panel, Black, 10A	P1, P2	Banana_AUT	Banana Socket	2
DF13-6P	HIROSE(HRS) DF13-6P-1.25H(20) Wire-To-Board Connector, Right Angle, DF13 Series, Surface Mount, Header, 6 Contacts, 1.25 mm	P3, P5	DF13-6	DF13-6P	2
HDR1X7V	Header, Plug, 2.54mm, 7 Pins, Straight	P9	HDR1X7_AUT	HDR1X7V	1
JST4	JST XA Header, 4-Pin, Pitch 2.5mm	P13, P15	XA-4	JST XA Header 4	2
5k1	Resistor, SMD, 0805	R1, R18	RES_0805	Resistor 0805	2
10k	Resistor, SMD, 0805	R2, R9, R12, R13, R14, R15, R16, R17	RES_0805	Resistor 0805	8
1k	Resistor, SMD, 1206	R3	RES_1206	Resistor 1206	1
120R	Resistor, SMD, 1206	R4	RES_1206	Resistor 1206	1
31k6	Resistor, SMD, 0805	R5	RES_0805	Resistor 0805	1
5k36	Resistor, SMD, 0805	R6	RES_0805	Resistor 0805	1
9k1	Resistor, SMD, 0805	R7	RES_0805	Resistor 0805	1
332k	Resistor, SMD, 0805	R8	RES_0805	Resistor 0805	1
0R	Resistor, SMD, 0805	R10, R11	RES_0805	Resistor 0805	2
EVQPF	PANASONIC EVQPF203M SWITCH, TACTILE, SPST-NO, 20mA, 15VDC	SW1	EVQPF203M	EVQPF	1
Terminal1	Terminal, Pad 70x70mil, Hole 32 mil	T1	PAD1_AUT	Terminal1	1
STM32C552RET6	Integrated Circuit	U1	LQFP40P900X900X145-64N	STM32C552RET6	1
2k	3362S Series, 1/4 " Square Potentiometer, 2k, 10%, 0.5 Watts, 1Turn	VR1	3362S	3362S-1-202LF	1
30k	3362S Series, 1/4 " Square Potentiometer, 2k, 10%, 0.5 Watts, 1Turn	VR2, VR3	3362S	3362S-1-202LF	2
24MHz	Crystals, 24MHz	X1	CX2520SB	CX2520SB24000D0FLJZZ	1